

Data Structures & Algorithms in JavaScript Handbook

Subject: JavaScript Core & Algorithms

=====

STUDYVAULT VERIFIED ACADEMIC NOTES & STUDY GUIDE

=====

1. ESSENTIAL DATA STRUCTURES IN JS

- Arrays: Contiguous memory storage with $O(1)$ random access.
- Hash Maps Object/Map: $O(1)$ key-value lookup performance.
- Linked Lists: Pointer-based node chains for fast $O(1)$ insertions/deletions.

2. BIG-O TIME COMPLEXITY ANALYSIS

- $O(1)$: Constant time | $O(\log N)$: Binary search | $O(N)$: Linear iteration.
- $O(N \log N)$: QuickSort & MergeSort | $O(N^2)$: Nested loops.